Sathira Silva

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Interests

Generative Diffusion | Vision-Language Models | Self-supervised Learning

Education

University of Peradeniya, Sri Lanka **Nov. 2018 – Dec. 2023**
B.Sc. (Hons) in Computer Engineering *GPA: 3.60 / 4.00*
[Courses, Thesis] *Class: 2^{nd} Upper*

De Mazenod College **Jan. 2003 – August 2016**
G.C.E. Advanced Level *Physical Science Stream*

Experience

Research Engineer **May 2025 – Current**
Computer Vision Department — Principal Investigator: *Dr. Muhammad Haris Khan* *MBZUAI*

- Working on extending existing benchmarks for developmentally inspired pre-training of LVLMs with a focus on temporal understanding.

Research Assistant **June 2024 – April 2025**
Computer Vision Department — Principal Investigator: *Dr. Muhammad Haris Khan* *MBZUAI*

- Contributed to developing a SOTA solution (DPA) to unsupervised fine-tuning of VLMs (CLIP) using *novel dual prototypes*.
- Fabricated a *novel multi-crop-based self-training pipeline* for fine-grained unsupervised adaptation of CLIP using fine-grained interactions between the two modalities, outperforming the SOTA.
- Worked on a *novel method for uncovering CLIP's hidden fine-grained knowledge* for unsupervised adaptation of CLIP on fine-grained datasets.

Computer Vision Research Engineering Intern **Dec. 2022 – Mar. 2023**
Autonomous Vehicle R&D Division *Vega Innovations*

- Contributed to the integration of a transformer architecture called NEAT into an autonomous vehicle system, by reviewing the paper and understanding its internals.
- Developed real-time computer vision solutions for autonomous vehicles on high performance GPU inference embedded systems (Nvidia DRIVE PX2 / Jetson TX2).

Teaching Assistant: Programming Methodology (CO222) **May 2021 – Sep 2021**
Department of Computer Engineering *University of Peradeniya*

- Supervised weekly 2hr long online lab sessions.
- Created questions for online quizzes based on the C programming Language.
- One-on-one sessions with students to tutor them on the C programming language concepts.

Selected Achievements

ACES Coders v7.0 | An inter-university algorithmic programming competition organized by of UoP **2020**
Rank - 14 / 100+
Team Name: bitLasagna

IEEEXtreme 14.0 | 24-hour global algorithmic programming competition **2020**
Country Rank - 2, Global Rank - 68 / 7000+
Team Name: InterGreat

ICDS Mini Hackathon | An inter-university Data Science Hackathon **2021**
Rank - 5 / 100+
Team Name: bitLasagna

IEEEXtreme 16.0 | 24-hour global algorithmic programming competition **2022**
Country Rank - 27, Global Rank - 427 / 6373
Team Name: bitLasagna

ACES Coders v9.0 | An inter-university algorithmic programming competition organized by of UoP **2022**
Rank - 2 / 100+
Team Name: bitLasagna

Ongoing Projects

Infants Learn from Continuous Experiences, Not Static Images!

Index Terms: Infant Learning, Data-efficient LVLM Pre-training, SAYCam

- **Motivation:** Developmentally aligned pre-training offers a path toward more data-efficient and cognitively plausible models. Unlike human infants who learn from continuous experiences, most developmentally inspired pre-training methods for VLMs (CVCL, BabyVLM) rely on static image–text pairs (e.g., SAYCam), limiting temporal and causal understanding.
- **Limitations of Prior Work:** BabyVLM and CVCL improve alignment using static, child-directed data, but still lack temporal structure essential for modeling real-world sequences.
- **Current Work:** I am developing a developmentally plausible video benchmark to evaluate and train LVLMs on continuous multimodal input, aiming to improve temporal grounding and event-level understanding.

Publications

microCLIP: Unsupervised CLIP Adaptation via Coarse-Fine Token Fusion for Fine-Grained Image Classification **Aug. 2025**
In Under Review for AAAI 2026
Authors: Sathira Silva, Eman Ali*, Chetan Arora, Muhammad Haris Khan*

Towards Fine-Grained Adaptation of CLIP via a Self-Trained Alignment Score arXiv **July 2025**
In Under Review for WACV 2026
Authors: Eman Ali, Sathira Silva, Chetan Arora, Muhammad Haris Khan

DPA: Dual Prototypes Alignment for Unsupervised Adaptation of Vision-Language Models arXiv poster **Oct. 2024**
In 2024 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)
Authors: Eman Ali, Sathira Silva, Muhammad Haris Khan

S2TPVFormer: A Spatiotemporal Approach to Tri-Perspective Representation for 3D Semantic Occupancy Prediction arXiv poster **Dec. 2024**
In AAAI 2025 Machine Learning for Autonomous Driving (ML4AD) Workshop – *Best paper award*
Authors: Sathira Silva, Savindu Wannigama*, Gihan Jayatilake, Muhammad Haris Khan, Roshan Ragel*

Other Projects

Autonomous Vehicle Emulator System (Internship Project) **Dec. 2022**
Index Terms: Image Processing, Computer Vision, Autonomous Driving *Group*
Technologies: Python, OpenCV, PyTorch, ONNX

- As a prototype for inferencing various autonomous driving trajectory prediction neural networks deployed on GPU accelerated hardware, implemented an emulator system in collaboration with the Vega Innovations Autonomous Vehicle team.
- Implemented a neural network deployment pipeline by converting PyTorch saved models to ONNX and generating optimized computational graphs using Nvidia libraries.
- Used ArUco markers and OpenCV to implement a location tracking system to generate waypoints for the prototype RC car implemented by the Vega team.

Sobriety Detection using Mobile Phone Gyroscope Data | **Jan. 2022**
Index Terms: Time-Series Forecasting, Feature Engineering | *Group*
Technologies: Python, TensorFlow, Scikit-learn, NodeJS

- Analyzed gyroscope data by visualization using signal processing techniques.
- Data cleaning, preprocessing and feature extraction using various methods.
- Implemented machine learning and deep learning models to classify the data.
- Contributed to develop a Node server to collect and process the data.
- Contributed to develop a prototype mobile application to send the data to the server.

Technical Skills

Languages: C/C++, Python, Java, HTML/CSS, JavaScript, SQL

Developer Tools: Visual Studio, VS Code, Eclipse, Jupyter Notebook, Android Studio

Technologies/Frameworks: OpernMMLab, PyTorch, TensorFlow, Bash Scripting, GitHub, OpenCV, TensorFlow, ReactJS, NodeJS, Jekyll

Certifications

Natural Language Processing (hons) | **Jan. 2022**
HSE University | *Coursera*

Algorithms on Graphs | **July 2020**
University of California San Diego | *Coursera*

Data Structures | **June 2020**
University of California San Diego | *Coursera*

Convolutional Neural Networks | **Feb. 2020**
DeepLearning.AI | *Coursera*

Neural Networks and Deep Learning | **Jan. 2020**
DeepLearning.AI | *Coursera*

Relevant Coursework

Data Structures & Algorithms	Operating Systems
Software Methodology	Computer Architecture
Image Processing	Programming Methodology
Artificial Intelligence	Discrete Mathematics
Networking and Web Application Design	Probability and Statistics

References

Dr. Muhammad Haris Khan

Assistant Professor,
Department of Computer Vision,
Mohammed bin Zayed University of AI,
United Arab Emirates

Prof. Roshan G. Ragel

Head of Department,
Department of Computer Engineering,
Faculty of Engineering,
University of Peradeniya, Sri Lanka